The Systems Behind Modern AI

Why and how does hardware designed for 3D games power modern AI

Sam Ly

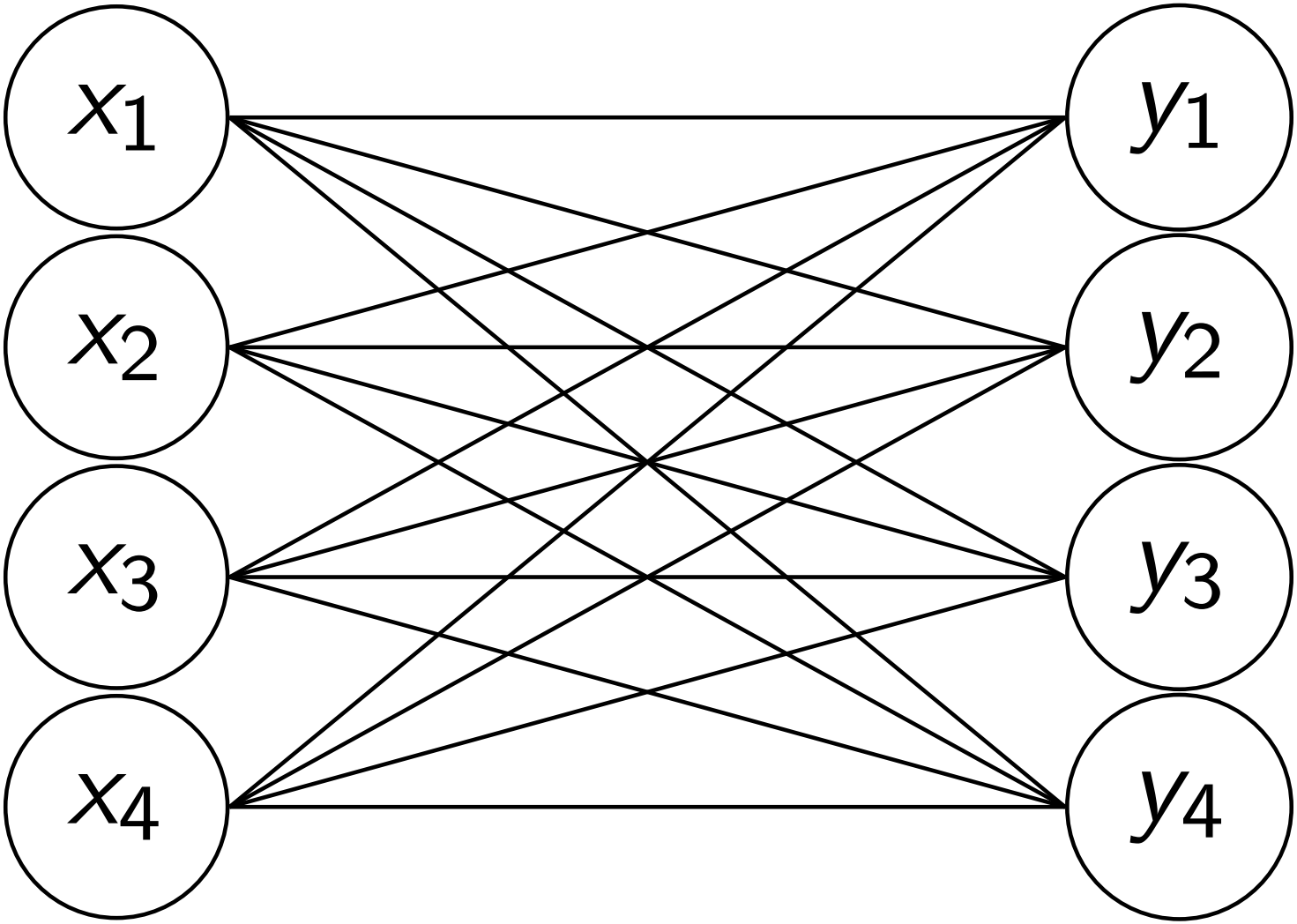
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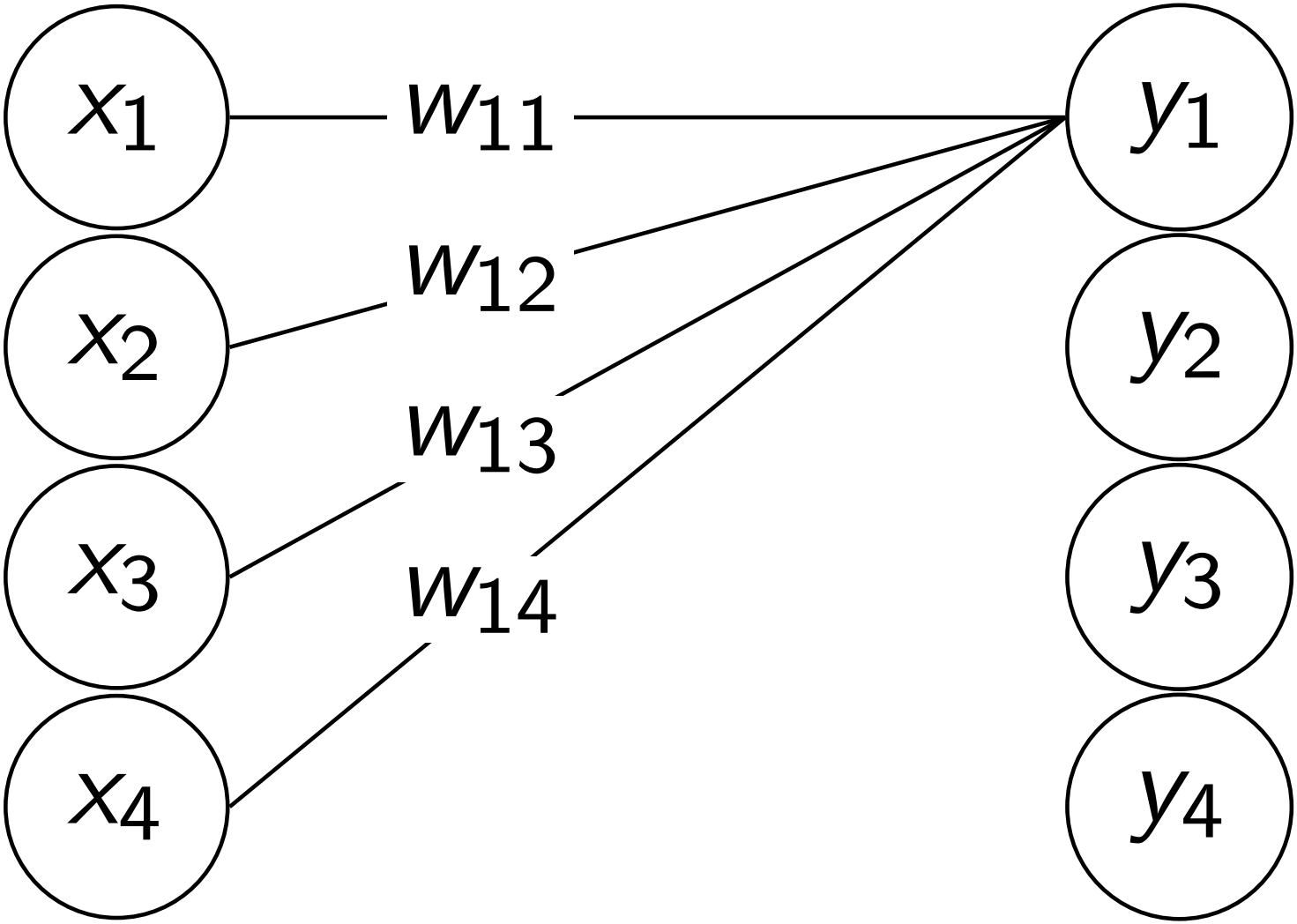
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Basic Neural Net (MLP/FFN)

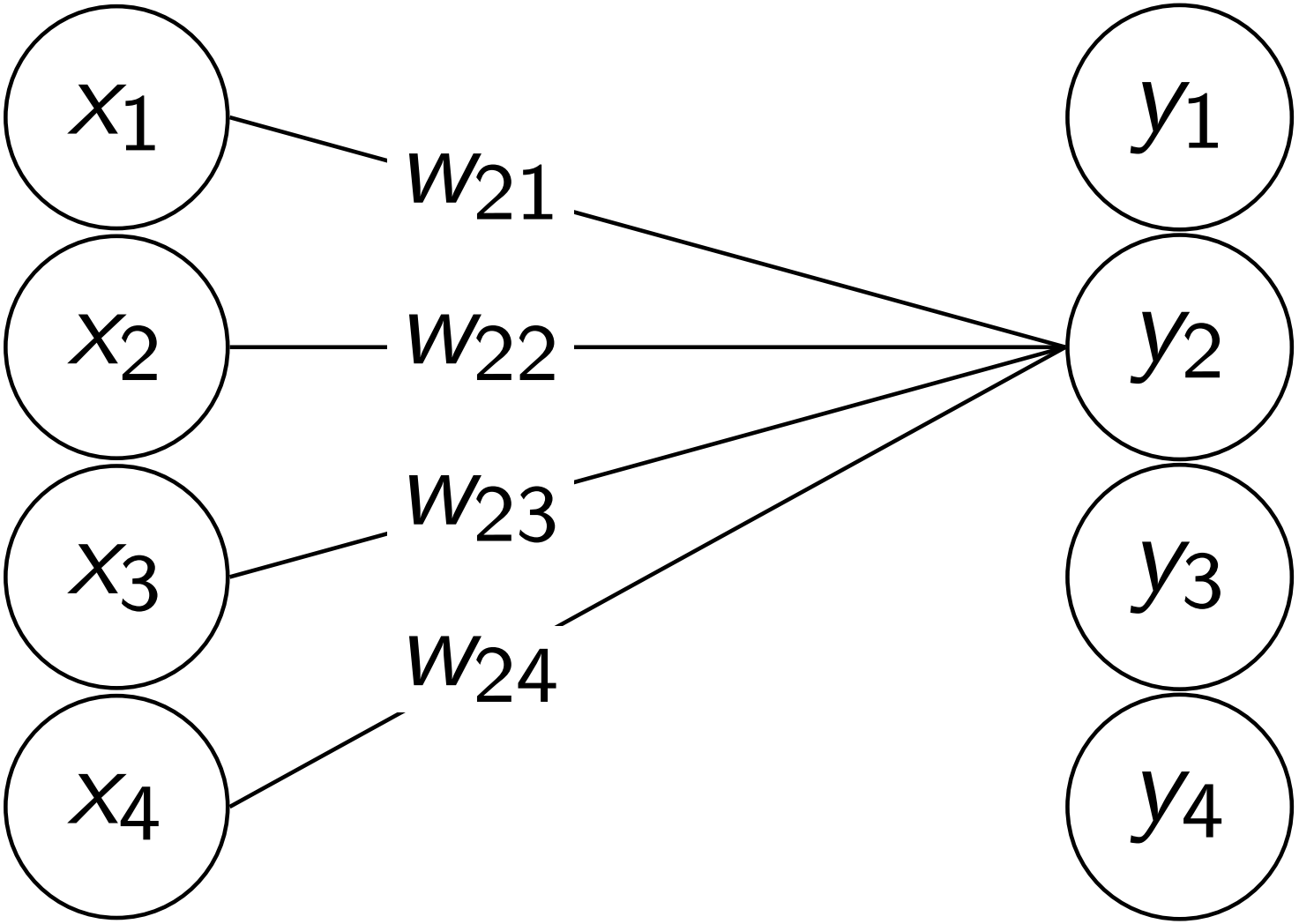


Basic Neural Net (MLP/FFN)



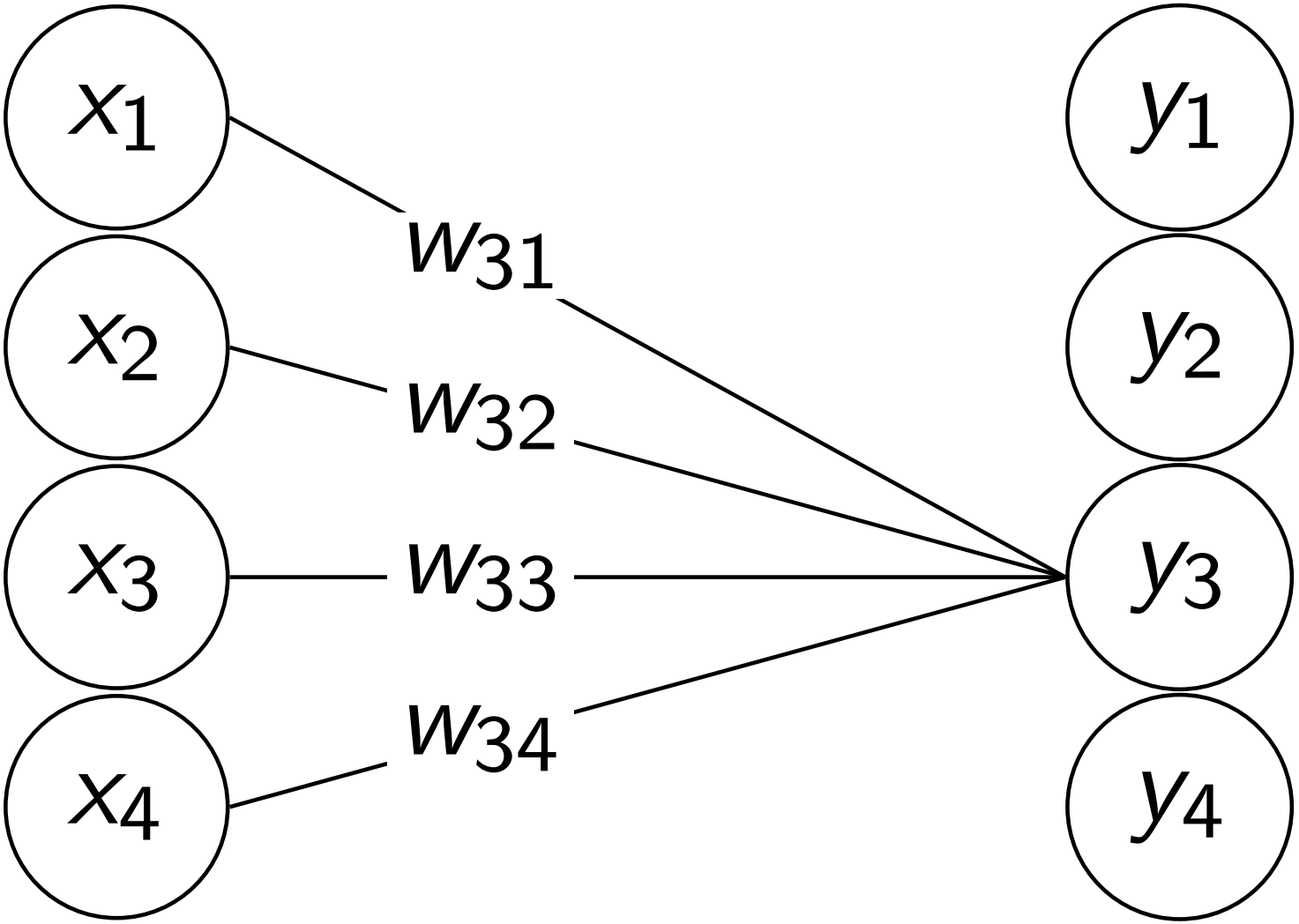
$$y_1 = x_1 w_{11} + x_2 w_{12} + x_3 w_{13} + x_4 w_{14}$$

Basic Neural Net (MLP/FFN)



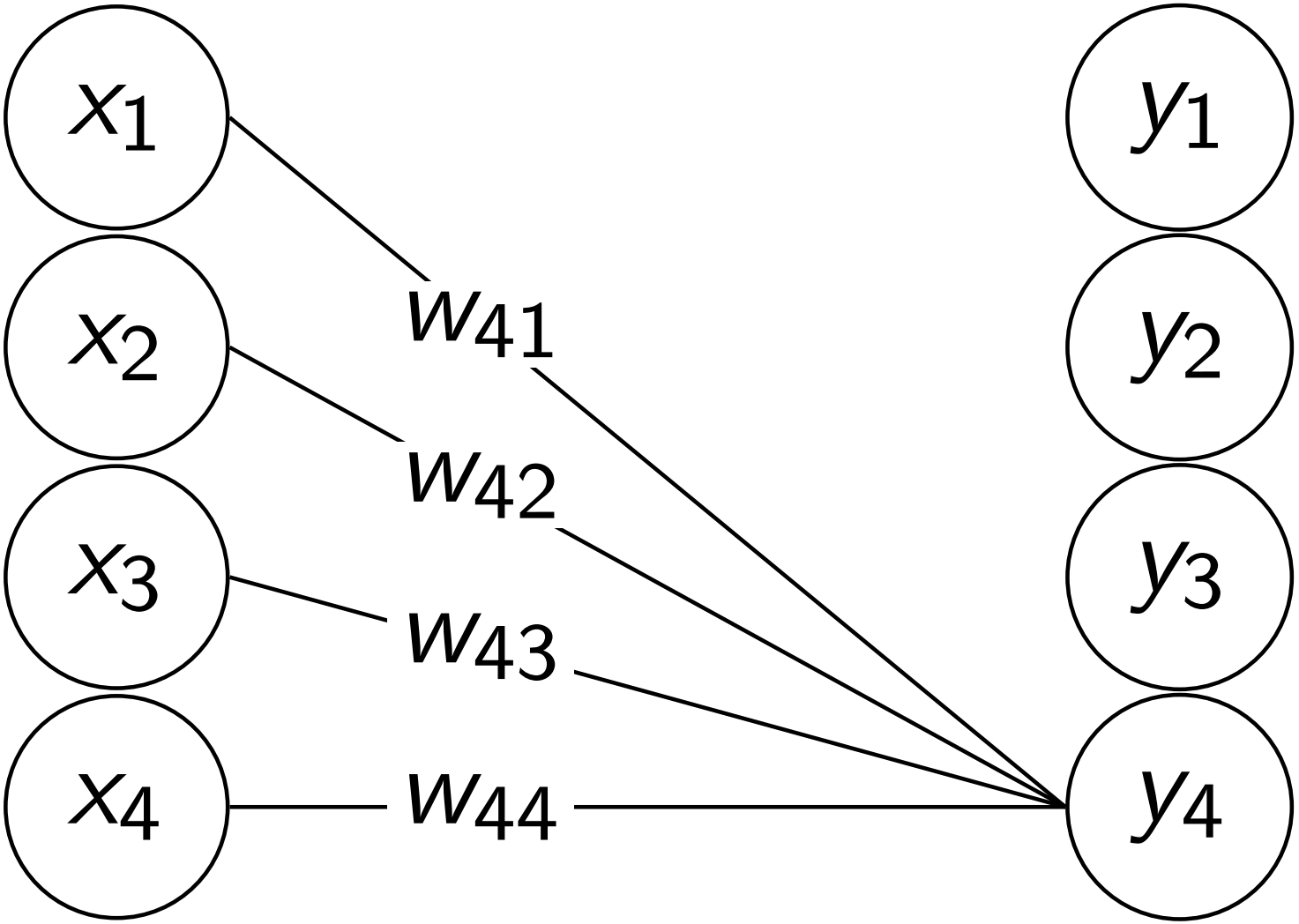
$$y_2 = x_1 w_{21} + x_2 w_{22} + x_3 w_{23} + x_4 w_{24}$$

Basic Neural Net (MLP/FFN)



$$y_3 = x_1 w_{31} + x_2 w_{32} + x_3 w_{33} + x_4 w_{34}$$

Basic Neural Net (MLP/FFN)



$$y_4 = x_1 w_{41} + x_2 w_{42} + x_3 w_{43} + x_4 w_{44}$$

Linear Algebra Interpretation

Here, notice that for each y_i, we are taking a weighted sum of the x elements. That is,

$$y_i = x_1 w_{i1} + x_2 w_{i2} + x_3 w_{i3} + x_4 w_{i4}.$$

Arranging the y_i into a vector, we get

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ w_{21} & w_{22} & w_{23} & w_{24} \\ w_{31} & w_{32} & w_{33} & w_{34} \\ w_{41} & w_{42} & w_{43} & w_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

Linear Algebra Interpretation

We can generalize the number of x's and the number of y's.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1} & w_{m2} & \cdots & w_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

We say, n is the *input dimension* and m is the *output dimension*.

Computing with multiple x vectors.

Suppose we have multiple x vectors, for which we want to pass through our neural network. We can stack the multiple x vectors in the **batch**, which we compute with a matrix multiplication.

$$\begin{bmatrix} y_1^{(1)} & y_1^{(2)} & \cdots & y_1^{(B)} \\ y_2^{(1)} & y_2^{(2)} & \cdots & y_2^{(B)} \\ \vdots & \vdots & \ddots & \vdots \\ y_m^{(1)} & y_m^{(2)} & \cdots & y_m^{(B)} \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1} & w_{m2} & \cdots & w_{mn} \end{bmatrix} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(B)} \\ x_2^{(1)} & x_2^{(2)} & \cdots & x_2^{(B)} \\ \vdots & \vdots & \ddots & \vdots \\ x_n^{(1)} & x_n^{(2)} & \cdots & x_n^{(B)} \end{bmatrix}.$$

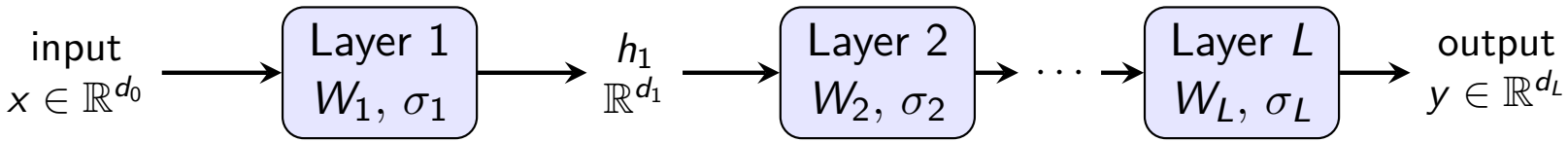
We write this succinctly as $Y = WX$, where $Y \in \mathbb{R}^{m \times B}$, $X \in \mathbb{R}^{n \times B}$, and $W \in \mathbb{R}^{m \times n}$.

Yippee!

We have successfully mapped the batched computation of a **linear layer** a matrix multiplication!

Composing Layers

We can view W as a function that takes in an n-dimensional vector and maps it to an m-dimensional vector. $W : \mathbb{R}^n \to \mathbb{R}^m$. We create deep neural networks by *composing* these transformations, along with **nonlinearities** σ_i.

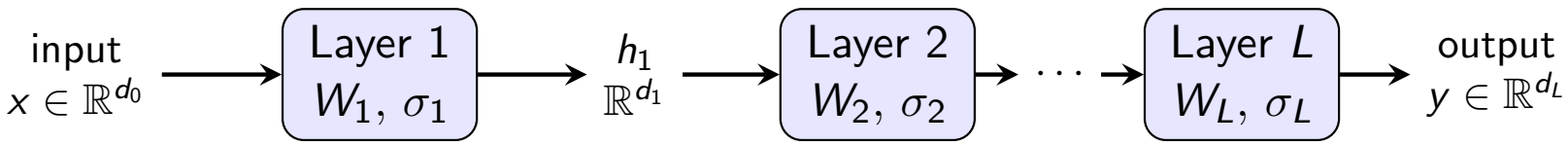


$$h_0 = x, \qquad h_\ell = \sigma_\ell(W_\ell h_{\ell-1}), \qquad y = h_L$$

Remark

Notice that this figure doesn't show the *batch* dimension. However, if you work through the dimension mappings, the *batch* dimension B will be unaffected by this composition.

Limitations for Neural Network Architecture



$$h_0 = x, \qquad h_\ell = \sigma_\ell(W_\ell h_{\ell-1}), \qquad y = h_L$$

Architectures such as **Transformers** and **CNNs** have connectivity patterns more complex than a simple composition of layers. Nevertheless, linear transformations remain a fundamental computational primitive in neural network architectures.

A fundamental building block

A linear transformation may seem like a narrow operation to study, but modern neural networks perform it repeatedly and at enormous scale. Its efficiency therefore has an outsized effect on training and inference performance. The rest of this presentation explores the hardware and techniques that make these operations efficient.

Takeaways

1. A linear layer applied to a batch of inputs can be expressed as a single matrix multiplication, $Y = WX$.
2. Deep networks compose these linear transformations with nonlinearities. More complex architectures change how layers are connected, but linear transformations remain a fundamental building block.
3. **Matrix multiplication therefore captures a large and reusable part of neural network computation.**

Why is this representation useful?

It separates what the network computes from how the hardware executes it. Instead of optimizing every architecture from scratch, we can focus on one common computational primitive. Next, we will examine how GPUs execute matrix multiplications efficiently.

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History of GPUs

- 1999: GeForce 256
- 2004: Brook Programming, more complex "rendering" pipelines, GP-GPU
- 2006: GeForce 8800, Compute Unified Device Architecture (CUDA)
- 2008: Fermi, General Purpose Scientific Compute becomes first class, L1/L2 caches
- 2010-2016: incremental improvements to power efficiency, programming models, and multi-GPU systems.
- 2017: Volta, AI-specialization: tensor cores designed around deep learning primitives
- 2022: Hopper, Transformer Engine, further specializing into AI
- 2026: Since 2022, we have only really seen incremental improvements.

Demand for 3D graphics

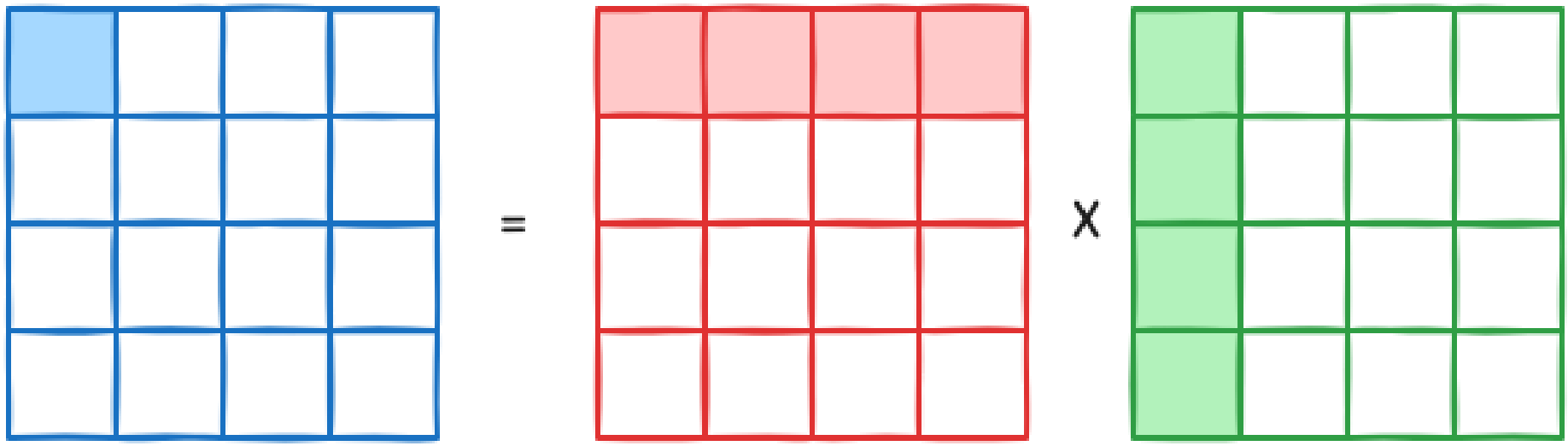
GPU's came about due to the increase in computational power needed to render 3D games in real time. This requires *millions* of FLOP's per second. CPU's could not keep up.

Parallelism

Rendering graphics has a nice property that we can take advantage of: **parallelism**.
Each pixel on the screen doesn't really care any of the other pixels. Furthermore, the computation that is required for each pixel isn't that complex either. The difficulty in rendering graphics is the volume of computations.

Computing a Matrix Multiplication

Let A and B be two 4×4 matrices. The matrix multiple $C = AB$ results in another 4×4 matrix.



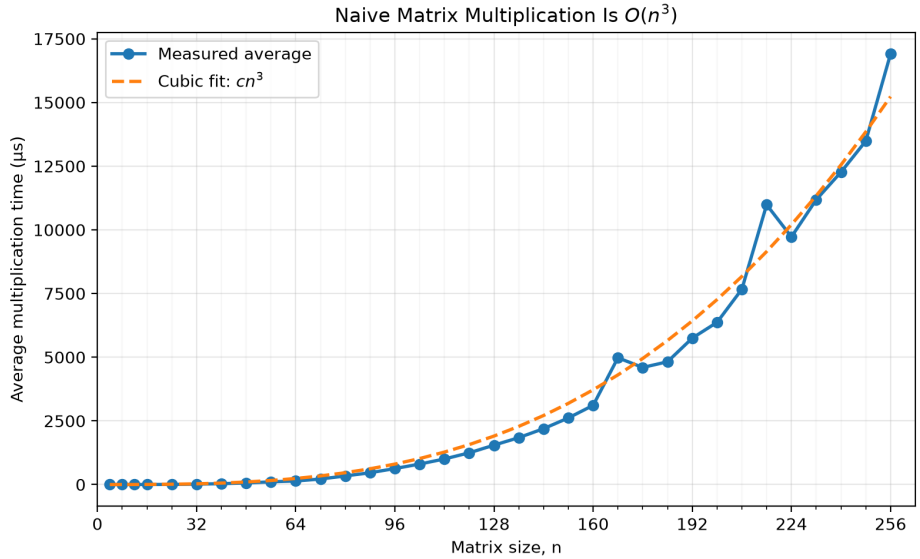
The operations are very simple, but the number of operations grow on the order of $\mathcal{O}(n^3)$, where n is the dimensions of the matrices.

Remark

Linear algebra and matrix multiplications show up everywhere in scientific computing, so computing them efficiently has always been in high demand.

Runtime of a matrix multiplication on a CPU

```
for (std::size_t i = 0; i < n; ++i) {
  for (std::size_t j = 0; j < n; ++j) {
    double sum = 0.0;
    for (std::size_t k = 0; k < n; ++k) {
      sum += a[i * n + k] * b[k * n + j];
    }
    c[i * n + j] = sum;
  }
}
```



Matrix Multiplication is Parallel

Each element of the output is independent. The results of C_{11} does not effect the results of C_{12}, or any other element.

Remark

If matrix multiplication is a **parallel** task, and GPU's were designed for the **parallel** task of rendering graphics, can we repurpose GPU's to do matrix multiplication?

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Remark

If matrix multiplication is a **parallel** task, and GPU's were designed for the **parallel** task of rendering graphics, can we repurpose GPU's to do matrix multiplication? (Yes.)

What shaped modern GPU architecture

Early GPUs like the GeForce 256 had graphics rendering primitives baked into the hardware design. (Separate vertex vs. pixel processors, etc.)
In 2004, researchers at Stanford created *Brook for GPUs*, which abused graphics APIs to emulate linear algebra operations.
NVIDIA saw the demand for general purpose parallel processing, starts shifting their hardware away from graphics-specific primitives, releasing the GeForce 8800 along with CUDA in 2006, replacing the graphics-specific hardware with a pool of unified compute.
In 2008, NVIDIA released the Fermi generation, making general scientific compute **first class**. Key memory designs show a deprioritization of graphics-specific workloads in favor of general parallel processing.

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Naive

Arithmetic Intensity, throughput, and the Roofline Model